



49-MILLIY SERTIFIKAT TESTI

1. $\underbrace{333 \dots 333}_{40 \text{ ta}} + \underbrace{666 \dots 666}_{40 \text{ ta}}$ ni hisoblang.

A) $\underbrace{999 \dots 999}_{80 \text{ ta}}$

B) $\underbrace{1\,000 \dots 000}_{39 \text{ ta}} - 1$

C) $\underbrace{999 \dots 999}_{40 \text{ ta}}$

D) $\underbrace{1\,000 \dots 000}_{41 \text{ ta}} - 1$

2. Hisoblang.

$$\frac{2^4 + 2^2 + 1}{2^2 - 2 + 1} + \frac{3^4 + 3^2 + 1}{3^2 - 3 + 1} + \frac{4^4 + 4^2 + 1}{4^2 - 4 + 1}$$

A) 35

B) 15

C) 41

D) -1

3. Ikkita nasos bir vaqtda ishlaganda hovuz 2 soat 55 minutda tozalanadi. Agar ulardan birinchisi bu ishnu ikkinchisiga qaraganda 2 soat tezroq bajarsa, birinchi nasos alohida ishlaganda hovuzni qancha vaqtda tozalashi mumkin?

A) 5

B) 7

C) 4

D) 8

4. Anvar o'z mashinasini 8000 dollarga sotmoqchi edi. Avval u narxni ketma-ket 20% va 25% ga oshirdi. So'ng hosil bo'lgan narxdan ketma-ket 25% va 20% chegirma qilgan bo'lsa, mashina oxir-oqibat qanchaga sotildi?

A) 7300\$

B) 8000\$

C) 7800\$

D) 7200\$



5. Agar $\frac{10!}{7^3-7} = 3^a \cdot 4^b \cdot 5^c$ bo'lsa, $a \cdot b - c$ ni hisoblang.

- A) 1
- B) 2
- C) 3
- D) 4

6. Hisoblang.

$$2 \cdot \frac{\sqrt{6+2\sqrt{5}}}{\sqrt{6-2\sqrt{5}}} - \frac{\sqrt{7+\sqrt{5}}}{\sqrt{7-\sqrt{5}}} \cdot 2\sqrt{11}$$

- A) -4
- B) 1
- C) $-\frac{\sqrt{5}+11}{2}$
- D) $\frac{\sqrt{5}-1}{2}$

7. Hisoblang.

$$\sqrt[3]{\frac{1}{3}(\sqrt[3]{2}-1)(\sqrt[3]{2}+1)^3}$$

- A) $\frac{1}{2}$
- B) 1
- C) $\frac{1}{\sqrt[3]{3}}$
- D) $\sqrt[3]{3}$

8. Arifmetik progressiyaning dastlabki to'rtta hadi yig'indisi 16 ga, keyingi oltita hadining yig'indisi 84 ga teng. Progressiyaning birinchi hadini toping.

- A) 0
- B) 1
- C) 2
- D) 5

9. Geometrik progressiyada $b_7 - b_4 = -216$, $b_5 - b_4 = -72$ va $S_n = 1023$ bo'lsa, n ning qiymatini toping.

- A) 8
- B) 9
- C) 10
- D) 12



10. Agar $xyz = 1$ bo'lsa, $\frac{1}{1+x+xy} + \frac{1}{1+y+yz} + \frac{1}{1+z+zx}$ ning qiymatini toping.

- A) 1
- B) 1,5
- C) 2
- D) 3

11. Agar $a = 19,25$ bo'lsa, $\left(\frac{a+2}{a-2}\right) : \left(\frac{6a}{a^3-8} + \frac{2a}{a^2+2a+4} + \frac{1}{2-a}\right) - \frac{4a+4}{a-2}$ ifodaning qiymatini toping.

- A) 1
- B) 12,25
- C) 4
- D) 19,25

12. Soddalashtiring.

$$\left(\frac{\sin 2x}{2} + \frac{\sin^2 x}{\frac{1}{\operatorname{tg} x} + \frac{1}{\sin x}} \right) : \cos x$$

- A) $\sin x$
- B) $\cos x$
- C) $\operatorname{tg} x$
- D) $\operatorname{ctg} x$

13. Hisoblang.

$$\frac{2 \cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$$

- A) 1
- B) 2
- C) $\sqrt{3}$
- D) 0



14. Tengsizlikni yeching.

$$2^{2x+6} + 2^{x+7} - 17 > 0$$

- A) $(-3; \infty)$
- B) $(0; \infty)$
- C) $(-3; 0)$
- D) $(3; \infty)$

15. Tenglamaning haqiqiy ildizlari ko'paytmagini toping.

$$\log_3 x \cdot (\log_9 x + 7) = 12 \cdot (2 + \log_9 x)$$

- A) 1
- B) $\frac{1}{3}$
- C) $\frac{1}{9}$
- D) $\frac{1}{27}$

16. Tenglamaning haqiqiy ildizlar ko'paytmasini toping.

$$x^2 + 3x + |x + 3| = 0$$

- A) -2
- B) 3
- C) -4
- D) 2

17. a ning qanday qiymatlarida $x^2 + 2ax + 2a^2 + 4a + 3 = 0$ tenglama ildizlari kvadratlar yig'indisi eng katta bo'ladi?

- A) -3
- B) 18
- C) 3
- D) -18

18. $|3 - 5x| - |1 - x| \leq 3$ tengsizlikni nechta butun son qanoatlantiradi?

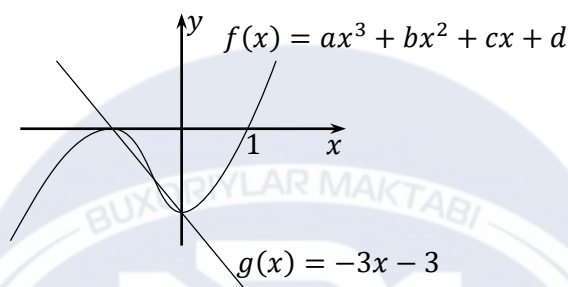
- A) 4 ta
- B) 1 ta
- C) 3 ta
- D) 2 ta

19. Tengsizlikni yeching. $3\sqrt{6+2x-4x^2} + 2 > 8x$

- A) $[-1; 0,25)$
- B) $[-1; 1)$
- C) $[0,25; 1)$
- D) $(-0,5; 1)$

20. Rasmda $f(x) = ax^3 + bx^2 + cx + d$ va $g(x) = -3x - 3$ funksiya grafiklari tasvirlangan. Funksiya grafiklaridan foydalanib $f(4)$ ning qiymatini toping.

- A) 200
- B) 215
- C) 225
- D) 250



21. Aniq integralni hisoblang.

$$\int_1^2 \frac{x+2}{\sqrt{x^2+4x+1}} dx$$

- A) $\sqrt{13} - \sqrt{6}$
- B) $\sqrt{13} + \sqrt{6}$
- C) 2
- D) 1

22. $f(x) = \cos(ax) \cdot \ln(x + \sqrt{1+x^2})$ funksiya berilgan. Agar $f'(2025) = 2026$ bo'lsa, $f'(-2025) + 2026$ ni hisoblang.

- A) 4052
- B) 4050
- C) 2026
- D) 0



23. Aylanada kesishuvchi ikkita AB va CD vatarlar o'tkazilgan. Agar $AB \perp CD$, $|AD| = 3$, $|BC| = 4$ bo'lsa, aylananing diametri uzunligini toping.

- A) $\sqrt{7}$
- B) 5
- C) 1
- D) $\sqrt{12}$

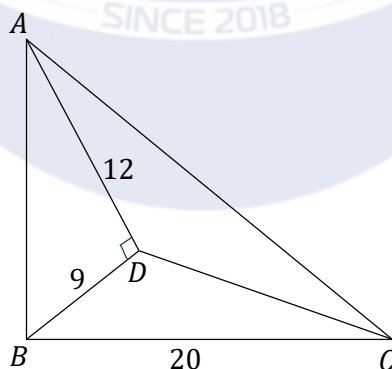
24. Quyidagilardan nechtasi toq funksiya?

- 1. $y = 3x^5 + x^3$
- 2. $y = (0,25)^x + (0,25)^{-x}$
- 3. $y = \begin{cases} x, & x < 0 \\ -x, & x \geq 0 \end{cases}$
- 4. $y = |x| - 1$
- 5. $y = \frac{x^4 - 2x^2}{3x}$
- 6. $y = \sqrt{3 - x^2 - 2x}$

- A) 2
- B) 3
- C) 4
- D) 1

25. ABC to'g'ri burchakli uchburchakda $AD \perp BD$, $AD = 12$, $BD = 9$, $BC = 20$ bo'lsa, ADC uchburchak yuzini hisoblang.

- A) 54
- B) 108
- C) 150
- D) 42



26. ABC uchburchakda B uchidan AC tomonga BM mediana va BH balandlik o'tkazilgan. Bunda $AB = \sqrt{5}$, $BM = 2\sqrt{2}$, $BH = 2$, $\angle ABC + \angle ACB < \frac{\pi}{2}$ bo'lsa, BC tomon uzunligini toping.

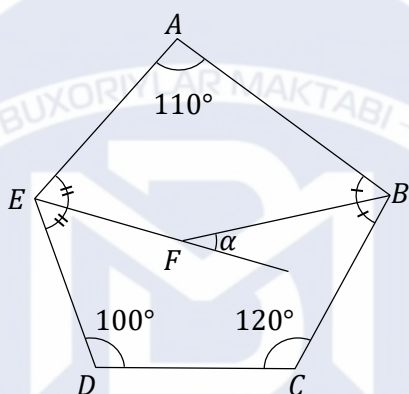
- A) 1
- B) $\sqrt{13}$
- C) 2
- D) $\sqrt{15}$

27. $ABCD$ trapetsiyada AD va BC tomonlari o'zaro parallel. CD yon tomonning o'rtasidan olingan K nuqtadan AB yon tomonga bo'lgan masofa 4 ga va $AB = 5$ bo'lsa, $ABCD$ trapetsiya yuzini hisoblang.

- A) 9
- B) $2\sqrt{5}$
- C) 20
- D) $\sqrt{5}$

28. $ABCDE$ beshburchakning $\angle EDC = 100^\circ$, $\angle EAB = 110^\circ$ va $\angle BCD = 120^\circ$ ga teng. Agar E va B uchlaridan chiqqan bissektrisalar F nuqtada kesishsa, α burchakni toping.

- A) 35°
- B) 40°
- C) 45°
- D) 50°

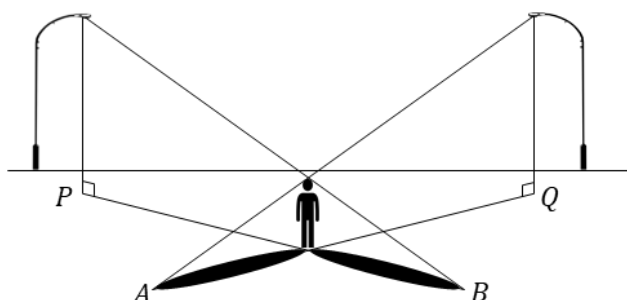


29. $DABC$ tetraedrda M, N, K va P nuqtalar mos ravishda DC, BC, AB va DA qirralarining o'rtalari. Tetraedrning qirrasi 4 ga teng bo'lsa, $\overrightarrow{MN} \cdot \overrightarrow{PK} + \overrightarrow{AB} \cdot \overrightarrow{BC}$ ni hisoblang.

- A) 2
- B) -4
- C) 4
- D) -8

30. Ikki dona ustun bor, har bir ustunning tepasida chiroq o'rnatilgan. Chiroqlarning yer sathidan balandligi 7 m. Ustunlar orasida bir bola turibdi, uning bo'yi 1,40 m. Har bir chiroq boladan yerga soya tushiradi. Har bir soyaning uzunligi 2,80 m. Soyalar yerda mos ravishda A va B nuqtalarda tugaydi. Soyalar uchlari orasidagi masofa $AB = 6$ m. Ikki ustun orasidagi masofa qancha?

- A) 12 m
- B) 32 m
- C) 48 m
- D) 24 m





31. To'plamlar Agar $A = \{(x, y) | y = x^2; x, y \in R\}$ va $B = \{(x, y) | y = 2x + 3; x, y \in R\}$ bo'lsa,

$A \cap B$ ni toping.

- A) $\{(1; 4); (-3; 9)\}$
- B) $\{(-1; 1); (3; 9)\}$
- C) $\{(-1; 1); (-3; 9)\}$
- D) $\{(1; 1); (3; 9)\}$

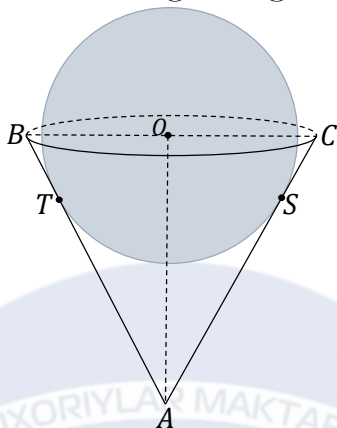
32. Tramvay bileti 0 dan 9 gacha raqamlar bilan belgilangan 6 xonali sondan iborat. Faqat 5 ta raqami bir xil son bo'ladigan biletlar soni nechta.

- A) 360
- B) 450
- C) 540
- D) 720



Topshiriqlar (33-35) va javob variantlari (A-F) ni o'zaro moslashtiring.

Konus shaklidagi vafliga markazi O bo'lgan shar shaklidagi muzqaymoq joylashtirilgan. Shar shaklidagi muzqaymoq konus shaklidagi vafliga T va S nuqtalarda urinadi. BAC muntazam uchburchak va O nuqta BC diametrda yotadi. Agar muzqaymoq sharining radiusi 4 cm ga teng bo'lsa, ($\pi = 3$ deb olinsin)



33. Konus asosi radiusining yarmini toping.

34. Konus to'la sirtini (cm^2) toping.

35. Muzqaymoqning yarmi konus ichiga erib tushsa, erigan muzqaymoqning A nuqtaga nisbatan chuqurligini (cm) toping.

A) $\frac{4\sqrt{3}}{3}$

B) $\frac{16}{\sqrt[3]{3}}$

C) 32

D) $4\sqrt[3]{6}$

E) $\frac{4\sqrt[3]{12}}{27}$

F) 128



36. Tenglamani yeching.

$$4(2^{x-1-2x^2} + 1)^2 + 3(2^{x-2x^2} - 3)(2^{x-2x^2} + 1) + 5 \cdot 2^{x+1-2x^2} = 0$$

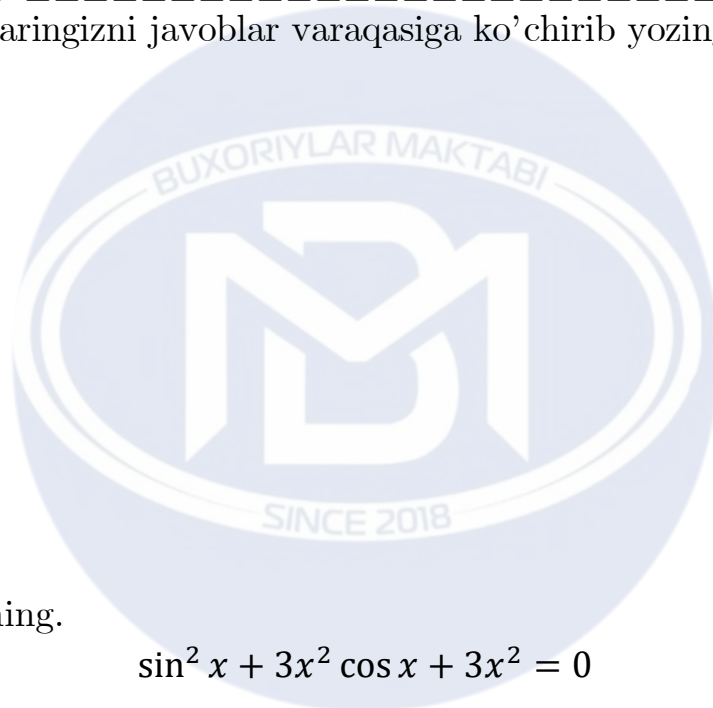
a) Tenglama nechta haqiqiy ildizga ega?

Javob: a) _____

b) Tenglamaning barcha haqiqiy ildizlari yig'indisini toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.



37. Tenglamani yeching.

$$\sin^2 x + 3x^2 \cos x + 3x^2 = 0$$

a) Tenglama $[0; 3\pi]$ oraliqda nechta ildizga ega?

Javob: a) _____

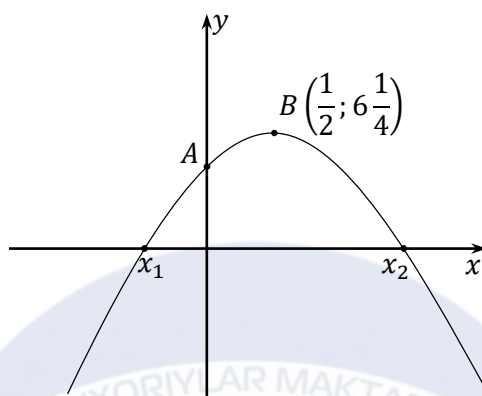
b) $\left[-\frac{\pi}{2}; \frac{5\pi}{2}\right]$ oraliqdagi yechimlari yig'indisini hisoblang.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.



38. Quyidagi chizmada $f(x) = ax^2 + bx + 6$ funksiyaning grafigi tasvirlangan. Parabola uchi $B\left(\frac{1}{2}; 6\frac{1}{4}\right)$ nuqtada joylashgan bo'lib, $f(x)$ funksiya grafigi Oy o'qini A nuqtada, Ox o'qini esa absissalari x_1 va x_2 ($x_1 < x_2$) bo'lgan nuqtalarda kesib o'tgan.



a) $\frac{x_2}{x_1}$ ni toping.

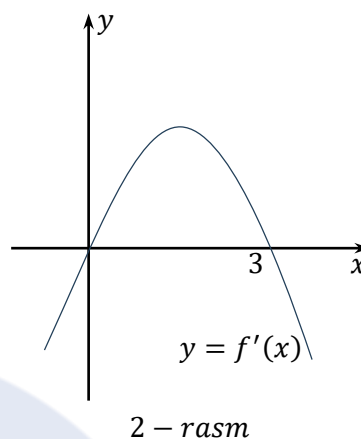
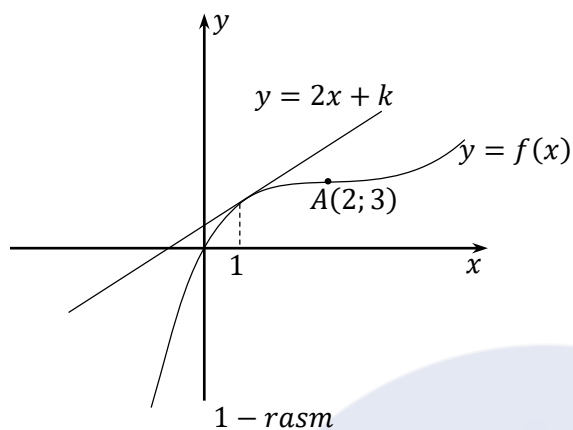
Javob a) _____

b) A va B nuqtalar orasidagi masofani toping.

Javob b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.

39. 1-rasmda $A(2; 3)$ nuqtadan o'tuvchi $y = f(x)$ egri chizig'iga $x = 1$ nuqtada o'tkazilgan urinma $y = 2x + k$ tasvirlangan. 2-rasmda esa $f(x)$ funksiyaning hosilasi $y = f'(x)$ grafigi bo'lgan parabola tasvirlangan. Agar $f^{-1}(2x - 1) = g(x)$ bo'lsa,



a) $g'(2)$ ni toping.

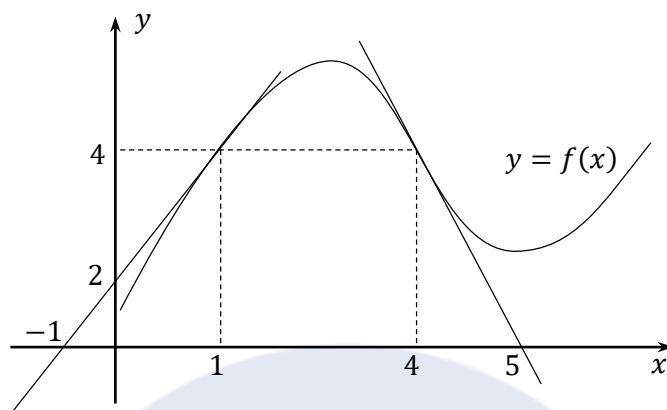
Javob: a) _____

b) $f'(2)$ ni toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.

40. $y = f(x)$ funksiyaga absissasi $x = 1$ va $x = 4$ nuqtalardan ikkita $y_1 = k_1x + b_1$ va $y_2 = k_2x + b_2$ urinmalar o'tkazilgan. Rasmdagi ma'lumotlar asosida quyidagilarni toping.



a) $\frac{k_1 \cdot k_2}{b_1 \cdot b_2}$ ning qiymatini toping.

Javob: a) _____

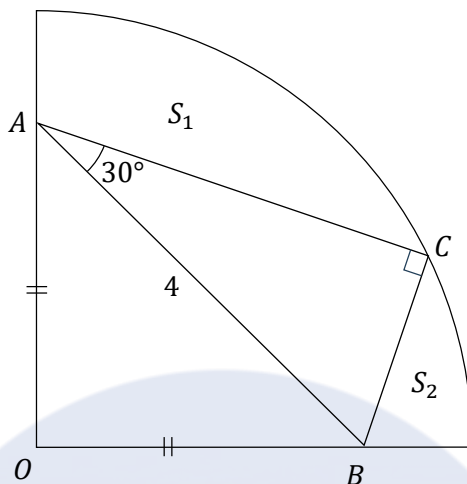
b) $\int_1^4 f''(x) f'(x) dx$ ning qiymatini toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.



41. O markazli chorak aylanaga ABC to'g'ri burchakli uchburchak rasmdagi ko'rinishda joylashtirildi. Agar $AO = OB$, $AB = 4$ va $\angle BAC = 30^\circ$ bo'lsa, ($\pi = 3$ deb oling)



a) Chorak aylana yuzini toping.

Javob: a) _____

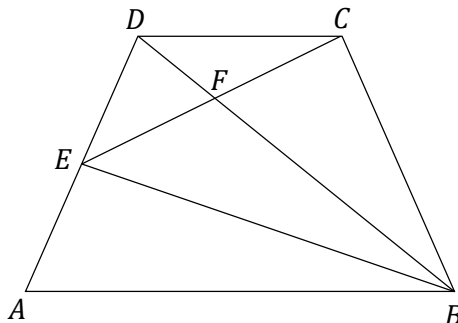
b) $S_1 + S_2$ ning qiymatini toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.



42. $ABCD$ trapetsiyada $DC \parallel AB$, $AE = DE$, DFC uchburchak yuzi 4 ga va EFB uchburchak yuzi 12 ga teng.



a) EDF uchburchak yuzini toping.

Javob: a) _____

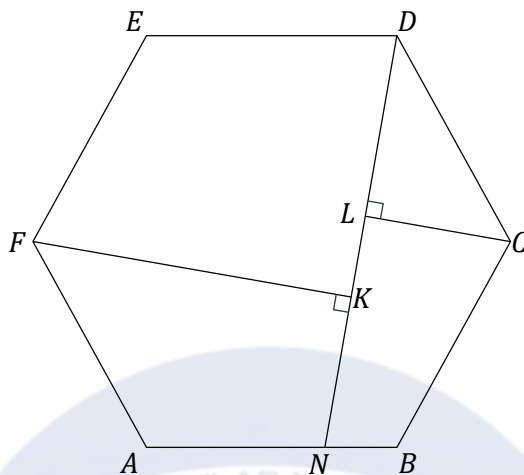
b) $\frac{S_{ABCE}}{S_{EDC}}$ ning qiymatini toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.



43. $ABCDEF$ muntazam oltiburchakning AB tomonidan N nuqta olindi. DN kesmaga F va C uchlariidan mos ravishda CL va FK perpendiykulyarlar tushirildi. Agar $AN = 3$ va $NB = 1$ bo'lsa,



a) LK ning qiymatini toping.

Javob: a) _____

b) LFD uchburchak yuzini toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.



44. Piramida asosi tomonlari 7, 8 va 9 bo'lgan uchburchakdan iborat. Piramidaning barcha qirralari asos tekisligi bilan 60° li burchak tashkil qilsa,

a) Piramidaning balandligini toping.

Javob: a) _____

b) Piramidaning hajmini toping.

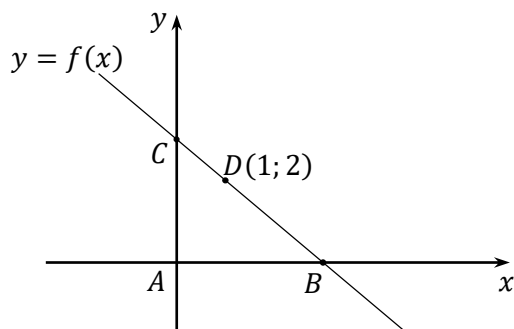
Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.





45. Rasmda $D(1; 2)$ nuqtadan o'tuvchi $y = f(x)$ to'g'ri chiziqli grafigi tasvirlangan. $y = f(x)$ funksiya Ox va Oy oqlarini mos ravishda B va C nuqtalarda kesib o'tadi. Agar ABC uchburchakning yuzi eng kichik bo'lsa,



a) BC kesma uzunligini toping.

Javob: a) _____

b) ABC uchburchak yuzini toping.

Javob: b) _____

Diqqat! Javoblaringizni javoblar varaqasiga ko'chirib yozing.